

Contribution

We present the **Rotational Unit of Memory (RUM)** — new fundamental Recurrent Neural Network (RNN) cell

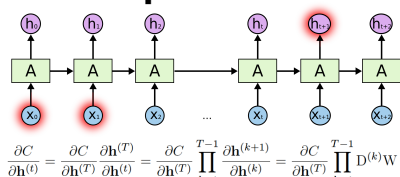
- 1) The **Rotation** operation is a phase-coding unit of associative memory, making RUM a flexible learner.
- 2) **Rotation** is naturally orthogonal, which mitigates the gradient vanishing/explosion problem.
- 3) We find that our architecture outperforms both LSTM/GRU and other state-of-the-art fundamental RNN cells in accuracy. Additionally, we obtain state-of-the-art results on **associative recall** and **character level language modeling** tasks.

Background & Motivation

Problem: inefficient memory encoding in conventional RNN cells

Gradient vanishing and explosion

Utilization of the hidden RNN state



$$h_t = \sigma(Wx_t + Ah_{t-1} + b)$$

known mitigation: associative memory (dynamic A)

known mitigations: batch norm., unitary init./“learnware” param.

$$A \rightarrow A_t$$

Our solution: rotations as associative memory and gradient stabilizers

- 1) $A_t \rightarrow O_t$ use phase space
- 2) $\|O_t h_{t-1}\| = \|h_{t-1}\|$ orthogonal help
- 3) $O_t \rightarrow R_t$ “firmware” rotations

Related works

- Restricted unitary space matrix parameterization [1]
- Hopfield net inspiration of associative memory [2]
- Firmware and learnfare structures [3]
- Rotation-like dynamic routing between capsules [4]

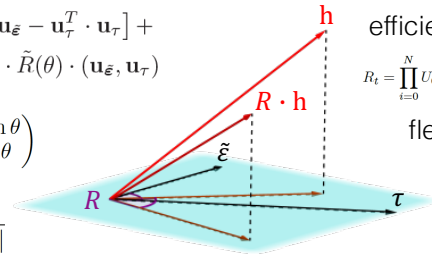
Model & Insight

Rotation: phase-coding

$$R(\tilde{\epsilon}, \tau) = [1 - \mathbf{u}_{\tilde{\epsilon}}^T \cdot \mathbf{u}_{\tilde{\epsilon}} - \mathbf{u}_{\tau}^T \cdot \mathbf{u}_{\tau}] + (\mathbf{u}_{\tilde{\epsilon}}, \mathbf{u}_{\tau})^T \cdot \tilde{R}(\theta) \cdot (\mathbf{u}_{\tilde{\epsilon}}, \mathbf{u}_{\tau})$$

$$\tilde{R}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$\theta = \arccos \frac{\mathbf{u}_a \cdot \mathbf{u}_b}{\|\mathbf{u}_a\| \|\mathbf{u}_b\|}$$

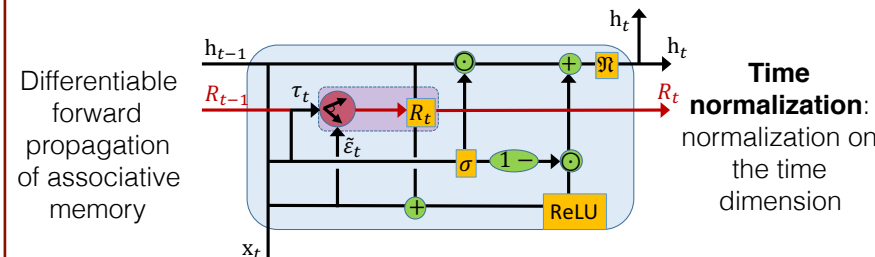


efficient phase-coding

flexible memory

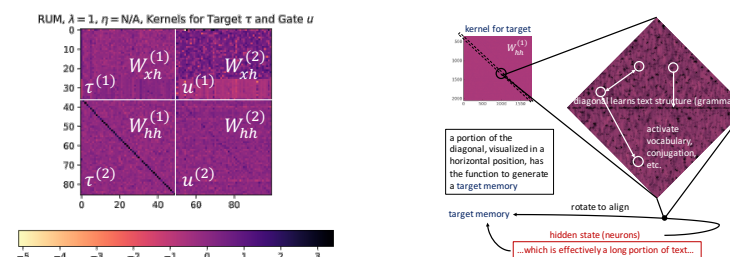
“firmware” operation on hidden state in phase space, no extra parameters!

Architecture of RUM



efficient implementation of the rotation as an associative memory, a new type of gates

Visualization of performance



temperature maps of weights on associative recall (left) and PTB (right)

Advantages

- 1) **Basic**: a new RNN cell with a new concept of gates
- 2) **Associative**: efficient phase-coding memorization
- 3) **Orthogonal**: more stable gradients through rotation, no need for bounded non-linearities
- 4) **Universal**: through 1), 2) and 3) RUM can serve as the building block of many high-level models

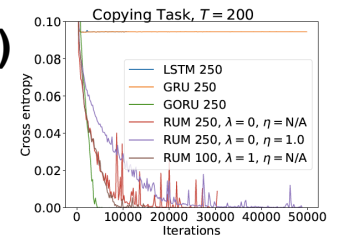
Experiments

We compare our model to LSTM/GRU and other basic RNN cells on accuracy of those skills: **memorize, recall, reason, predict**.

Copying task (memorize)

Task: read a long number, wait for time T and then output the number.

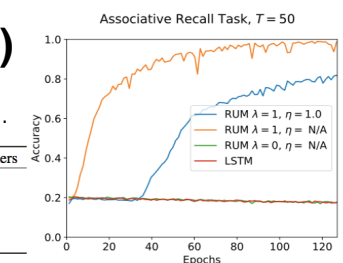
RUM learns fully while LSTM/GRU hit a random guessing baseline.



Associative recall (recall)

Task: read a sequence of length T, and recall what character follows after a “key”.

Model	Length T = 30	Length T = 50	# Parameters
LSTM	25.6%	20.5%	17k
GORU	21.8%	18.9%	13k
FW-LN (Ba et al. (2016a))	100%	20.8%	9k
WeiNet (Zhang & Zhou (2017))	100%	100%	22k
RUM (ours)	100%	100%	13k



RUM achieves 100% accuracy on the state-of-the-art T=50 with the least number of parameters.

bAbI question answering (reason)

Model	Test Accuracy (%)
LSTM (Weston et al. (2015))	49
GORU (Jing et al. (2017a))	60
MemN2N (Sukhbaatar et al. (2015))	86
RUM (ours)	73.2

Task: give simple answers to simple questions based on a given context.

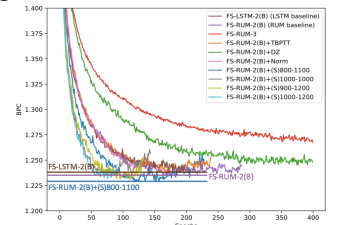
RUM has better accuracy than the other basic RNN cells.

Attention mechanism gives the state-of-the-art among all models.

Character level Penn Treebank (predict)

Task: guess next character while reading.

Model	BPC	# Parameters
Zoneout LSTM (Krueger et al. (2016))	1.27	—
RUM 2000 (ours)	1.28	8.9M
2 x RUM 1500 (ours)	1.26	16.4M
HM-LSTM (Chung et al. (2016))	1.24	—
HyperLSTM (Ha et al. (2016))	1.219	14.4M
NASCell (Zoph & V. Le (2016))	1.214	16.3M
FS-LSTM-4 (Mujika et al. (2017))	1.193	6.5M
FS-LSTM-2 (Mujika et al. (2017))	1.190	7.2M
FS-RUM-2 (ours)	1.189	11.2M



A high-level model based on RUM gives best test accuracy.

RUM outperforms LSTM/GRU on all the tasks, which test *diverse* properties, and is the state-of-the-art on *recall* and *predict*.

Reference & Code

- [1] M. Arjovsky, A. Shah, and Y. Bengio, “Unitary Evolution Recurrent Neural Networks,” *ICML 2016*
- [2] J. J. Hopfield, “Neural Networks and Physical Systems with Emergent Collective Computational Abilities,” *Proceedings of the national academy of sciences* 79(8): 2554-2558, 2016.
- [3] David Balduzzi and Muhammad Ghifary, “Strongly-Typed Recurrent Neural Networks”, *ICML 2016*.
- [4] Sara Sabour, Nicholas Frost and Geoffrey Hinton, “Dynamic Routing Between Capsules”, *NIPS 2017*.

Tensorflow: <https://github.com/jingli9111/RUM-Tensorflow>

PyTorch: <https://github.com/rdangovs/RUM-PyTorch>

Tensorpack: <https://github.com/rdangovs/RUM-Tensorpack>